Arbitrarily Long Gaps in the Squarefree Integers

OEIS A005117 November 12, 2026

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Abstract

For each positive integer L, we construct an explicit arithmetic progression of positive integers n for which none of $n, n+1, \dots, n+L-1$ is squarefree. Its natural density is a positive, explicitly computable number. Consequently, the gaps between successive terms of OEIS A005117 are unbounded. This is a known elementary Chinese-remainder construction, not a resolution of an open conjecture.

Definition and result

A positive integer is *squarefree* when no square of a prime divides it. OEIS A005117 lists the squarefree positive integers in increasing order, starting at 1 [1]. Denote them by $s_1 < s_2 < \cdots$.

Theorem 1. *Given any $L \geq 1$, there exist pairwise distinct primes $q_0, \dots, q_{L-1}$, a positive integer $M = \prod_{j=0}^{L-1} q_j^2$, and a residue class $r \pmod M$ such that*

$$q_j^2 \mid (n+j) \quad (0 \leq j < L) \tag{1}$$

for every positive integer $n \equiv r \pmod M$. Thus, the set of such starting points n has natural density exactly $1/M$, and every one of them begins a run of at least L consecutive non-squarefree integers.

Proof. Choose any L distinct primes $q_0, \dots, q_{L-1}$. Since the moduli q_j^2 are pairwise coprime, the Chinese remainder theorem supplies a unique class r modulo $M = \prod_{j=0}^{L-1} q_j^2$ satisfying

$$r \equiv -j \pmod{q_j^2}, \qquad j = 0, 1, \dots, L-1.$$

Every positive representative n of that class satisfies (1), so $n+j$ is divisible by a prime square and hence is not squarefree. This holds simultaneously for all j. Finally, for a fixed residue class, the number of positive $n \leq X$ in that class equals $X/M + O(1)$ as $X \to \infty$. Therefore the class has natural density $1/M > 0$. □

Corollary 1. *The differences $s_{k+1} - s_k$ are unbounded. More precisely, for each $L \geq 1$ there are infinitely many indices k satisfying $s_{k+1} - s_k \geq L+1$.*

Proof. By Theorem 1, there are infinitely many arbitrarily large starts n such that $[n, n+L-1]$ contains no squarefree integer. Infinitely many squarefree integers exist, because every prime is squarefree (Euclid's theorem). Thus the consecutive squarefree integers immediately bracketing this interval are well defined for n large, and their difference is at least $L+1$. For fixed L, infinitely many starts cannot all be contained in the same gap between successive squarefree numbers, since that would be a finite interval. Hence infinitely many *distinct* gaps satisfy the inequality. □

Example, scope, and priority

For $L = 3$ take $(q_0, q_1, q_2) = (2, 3, 5)$. One CRT solution is $n \equiv 548 \pmod{900}$; indeed $4 \mid 548$, $9 \mid 549$, and $25 \mid 550$. Thus all $n = 548 + 900t > 0$ begin three-term runs of non-squarefree integers.

The construction and unboundedness of squarefree gaps are classical and already documented [2, 3]. Quantitative upper bounds on how large these gaps can be are a separate, deeper research direction [4]. The result does not address unrelated prime-producing conjectures recorded in A005117, and makes no originality claim.

Verification disclosure

This proof was AI-authored and internally checked, including a finite test of the displayed CRT example. It has not been independently peer-reviewed; the cited sources establish the prior-known status of the central argument.

References

[1] OEIS Foundation, *A005117: Squarefree numbers*, `https://oeis.org/A005117` (accessed November 12, 2026).

[2] Mathematics Stack Exchange, *Is there an unlimited number of sequences of consecutive numbers with $\mu(n) = 0$ of any length?*, `https://math.stackexchange.com/questions/4057891` (2021).

[3] *Square-Primitive Gaps*, MIT math enrichment material (2016), `https://math.mit.edu/research/highschool/primes/materials/2016/conf/4-2%20Smith.pdf`.

[4] A. Kumchev, W. McCormick, N. McNew, A. Park, R. Scherr, and W. Ziehr, *Explicit bounds for large gaps between squarefree integers*, Journal of Number Theory (2023), `https://doi.org/10.1016/j.jnt.2023.07.003`.